In-gel protein staining with Coomassie Brilliant Blue

Coomassie Brilliant Blue is a pigment used to stain proteins in acrylamide or agarose gels with a detection limit of 10–20 ng per band. The staining is permanent and incompatible with Western blotting. However, Coomassie G-250 (not R-250) can also be used to visualize proteins on PVDF or nitrocellulose membranes. While silver staining SOP 0011 offers higher sensitivity, Coomassie-stained gels are compatible with downstream mass spectrometry.

Risk assessment

- Work with flammable liquids
- Work with corrosive liquids
- ▷ Wear gloves, safety glasses, lab coat
- ☐ Collect as HAZARDOUS WASTE



Reviewed: Jan 16, 2023

Procedures

›› *Coomassie Brilliant Blue staining*

- ☐ Incubation tray
- ☐ Lint-free wipe
- ☐ Coomassie destaining solution (20 mL / 1 rxn) R0123
- ☐ Coomassie staining solution (20 mL / 1 rxn) R0124

(1.) *Optional:* Place gel in a container with water. Microwave at 1 000 W for 1 min.

> ***This is why:*** This helps remove residual SDS and speeds up destaining. Don't boil; gels can melt or crack if microwaved dry.

(2.) Transfer gel into a tray with Coomassie staining solution. Rock gently at room temperature for 15 min. 15 min

(3.) Discard staining solution. Rinse briefly with water to remove excess dye.

(4.) Add Coomassie destaining solution. Rock at room temperature for 45 min. 45 min

(5.) Pour off the destaining solution into a collection bottle for reuse. Replace with water and add a lint-free wipe in the tray to soak up free dye. Destain overnight until background disappears.

Materials

Coomassie Brilliant Blue staining

Incubation tray

Lint-free wipe

Coomassie destaining solution R0123 (20 mL / 1 rxn)

Coomassie staining solution R0124 (20 mL / 1 rxn)

Recipes

Coomassie destaining solution

Amount	Ingredient		Stock	Final
1.2 L	Isopropyl alcohol	[67-63-0]	100%	30%
	Flammable liquid (H225)			
2.4 L	Water, reagent-grade	[7732-18-5]		
0.4 L	Acetic acid, glacial	[64-19-7]	99.6%	10%
	Skin corrosion (H314)			

Add concentrated acid slowly to the stirred water/alcohol mixture. DO NOT autoclave. ***Safety:*** Work inside a certified chemical fume hood. ***Note:*** The destaining solution can be reused several times and refreshed using activated charcoal. ***This is why:*** In this adaptation, 30% isopropyl alcohol or ethanol replaces the more toxic 45% methanol.

Coomassie destaining solution

30% Isopropyl alcohol, 10% Acetic acid



DANGER

Flammable liquid; Skin and serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0123

Coomassie staining solution

Amount	Ingredient		Stock	Final
1.0 g	Brilliant Blue G	[6104-58-1]	10 g/L	0.5 g/L
2.0 L	Coomassie destaining solution	◇ R0123		
	Flammable liquid (H225); Skin irritation (H315); Serious eye irritation (H319)			

Note: The staining solution can be used multiple times. ***Hint:*** Coomassie G-250 is more sensitive while Coomassie R-250 gives sharper bands at lower cost.

Coomassie staining solution



DANGER

Flammable liquid; Skin and serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0124

Troubleshooting

Coomassie Brilliant Blue staining

In Step 2:

- Faint or invisible bands
  - Increase staining time or protein loading. Ensure sufficient protein was loaded. The detection limit is approximately 10–20 ng per band.
  - Verify staining solution has not been exhausted from repeated reuse.
- Uneven staining across the gel
  - Ensure the gel is fully submerged and rocking evenly. Use sufficient volume of staining solution — at least 5 times the gel volume.

In Step 4:

- High background that does not clear with extended destaining
  - Replace destaining solution more frequently. Add a crumpled lint-free wipe or activated charcoal to absorb free dye.
  - If using R-250, switch to G-250, which gives lower background due to colloidal binding.

Change log

2020-06-12	Hai T. Dao	Initial protocol; Tom Muir lab "Coomassie stain and destain solution".
2023-01-16	Benjamin C. Buchmuller	Adaptation as SOP. Replaced methanol with less toxic isopropyl alcohol, from the Daniel Summerer lab "Coomassie staining" protocol.

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